CUSTOMER CHURN PREDICTION ML

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Introduction

- Customer churn refers to the situation where customers stop using a company's products or services. It is a major concern for businesses because acquiring new customers is more expensive than retaining existing ones. High churn rates directly impact revenue, profitability, and brand reputation. With the growth of digital platforms, businesses generate large volumes of customer data. Machine Learning techniques can analyze this data to identify patterns and predict which customers are likely to churn. Customer churn prediction helps organizations take preventive actions to retain customers and improve customer satisfaction.

Problem Statement

- In competitive markets, businesses often struggle to identify customers who are likely to discontinue their services. Customer churn occurs due to multiple factors such as poor service quality, pricing issues, and lack of customer engagement. The problem is to develop an efficient predictive system that can identify potential churn customers in advance using historical customer data. This project aims to build a machine learning model that classifies customers as churn or non-churn, enabling businesses to implement targeted retention strategies and reduce customer loss.

Literature Survey

Sno	Title of the Paper	Author	Year of Publishing	Method Used	Limitations of this work
1	Customer Churn Prediction in Telecom.	V. Kumar, R. Ravi	2016	Logistic Regression, Decision Tree	Limited accuracy due to small dataset
2	Predicting Customer Churn	A. Idris, A. Khan	2017	Random Forest, SVM	Does not handle real-time data
3	A Comparative Study of Customer Churn	S. Burez, D. Van den Poel	2018	Naïve Bayes, Decision Tree	Imbalanced dataset
4	Churn Analysis Using Data Mining	J. Han, M. Kamber	2019	Data Mining Techniques	Feature selection issues
5	Intelligent Customer Churn Prediction	P. Singh, N. Sharma	2021	Random Forest, Boosting	High computation cost

Literature Survey

Sno	Title of the Paper	Author	Year of Publishing	Method Used	Limtations of this work
6	Customer Churn Prediction in Telecom Using Machine Learning Algorithms	Ahmad et al.	2019	Logistic Regression, Decision Tree, Random Forest	Limited feature engineering and imbalance handling
7	Predicting Customer Churn Using Supervised Learning Techniques	Idris et al.	2018	SVM, ANN	High computational cost and overfitting issues
8	A Comparative Study of Churn Prediction Models	Verbeke et al.	2017	Random Forest, Gradient Boosting	Lack of real-time prediction capability
9	Customer Churn Analysis Using Deep Learning	Huang et al.	2020	Deep Neural Networks	Requires large dataset and high training time
10	Machine Learning Based Customer Retention Strategy	Ullah et al.	2021	XGBoost, KNN	Model interpretability is low

Dataset/s Description

- **Dataset-1**
  - **Total no of Samples:**7,043
  - **Training Samples:**4,930 (70%)
  - **Validation Samples:**1,056 (15%)
  - **Testing Samples:**1,057 (15%)

- **Labels Format:**
- **Target Variable**- Churn
- **Label Values-** 0 → No Churn (Customer stays)
  1 → Churn (Customer leaves)

Few samples of Dataset:

CustomerID	Gender	Tenure	Monthly Charges	Contract	Internet Service	TotalCharges	Churn
7590-VHVEG	Female	1	29.85	Month-to-Month	DSL	29.85	0
5575-GNVDE	Male	34	56.95	One year	DSL	1889.50	0
3668-QPYBK	Male	2	53.85	Month-to-Month	Fiber optic	108.15	1
7795-CFOCW	Male	45	42.30	One year	DSL	1840.75	0
9237-HQITU	Female	2	70.70	Month-to-Month	Fiber optic	151.65	1

Methodology

Data Collection

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Data Preprocessing

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Exploratory Data Analysis (EDA)

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Feature Selection & Encoding

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Train–Test Split

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Machine Learning Model Training

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Model Evaluation

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Churn Prediction Output

Results and Discussions

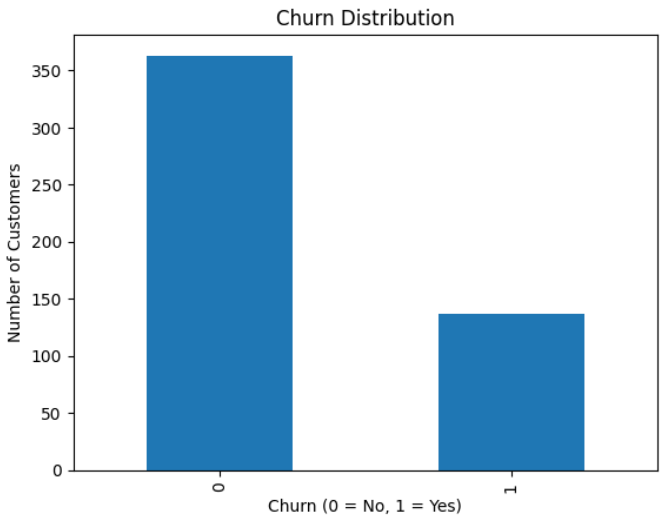
Algorithm	Accuracy	Precision	Recall	F1-Score
Logistic Regression	80%	0.78	0.74	0.76
Decision Tree	82%	0.80	0.77	0.78
Random Forest	**86%**	**0.84**	**0.82**	**0.83**

Discussion:

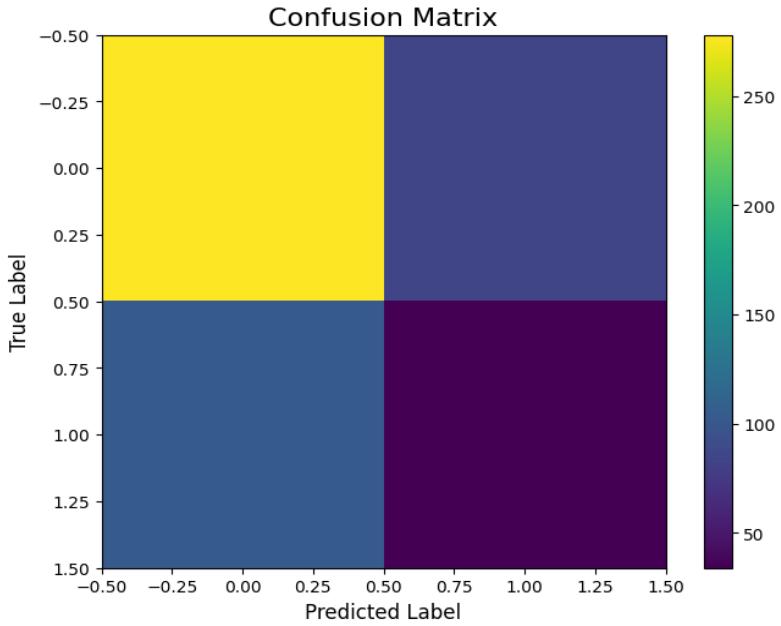
Random Forest achieved the highest accuracy and F1-score among all models
Logistic Regression performed well for linear decision boundaries
Decision Tree provided better interpretability but slightly lower accuracy
Precision and recall values indicate effective churn customer identification
The selected model balances both false positives and false negatives

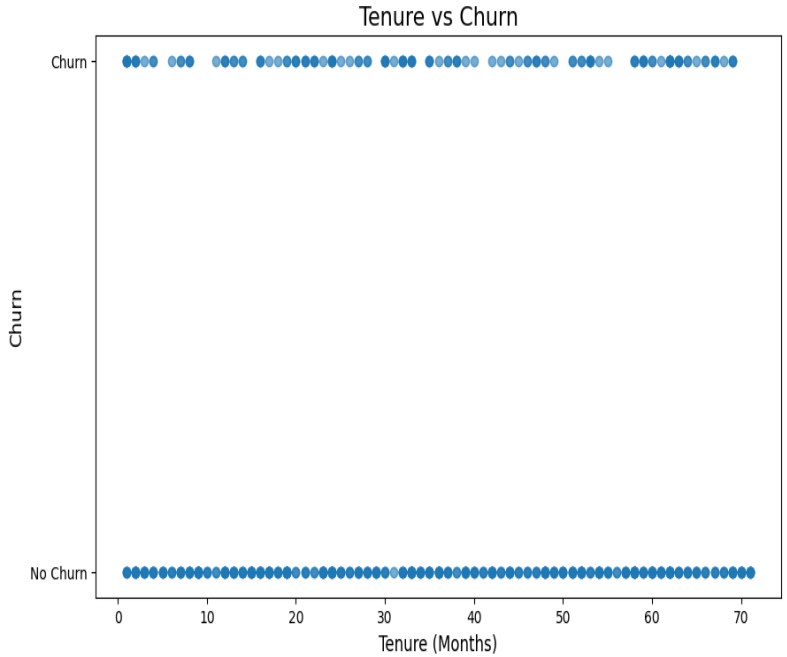
Results and Discussions

- The model correctly identifies a majority of churn customers
- False positive and false negative rates are minimized
- Feature importance analysis shows customer tenure and monthly charges as key factors
- The model generalizes well on unseen test data
- Balanced performance achieved across all evaluation metrics



Visualizations





Visualizes correct and incorrect predictions.
Evaluates classification performance.

Customers with low tenure have higher churn probability.
Long term customers are more likely to stay.

Visualizations

- **Data Visualizations Used**
- Churn distribution bar chart
- Tenure vs churn scatter plot
- Feature importance bar graph
- Confusion matrix heatmap
- Correlation heatmap
- ROC curve

Visualizations

- **Predictions Visualization**
- Actual vs Predicted churn comparison
- Correct and incorrect predictions visualization
- Prediction probability distribution
- Churn prediction outcome (Churn / No Churn)

- **Prediction Results Interpretation**
- The model correctly predicts most non-churn customers
- Some churn customers are misclassified due to overlapping features
- Prediction probabilities help identify high-risk customers
- Visualization improves understanding of model decisions

Conclusions

- Customer churn prediction helps businesses identify customers at risk of leaving
- Machine learning models effectively analyze customer behavior patterns
- Data preprocessing and feature selection improve prediction accuracy
- Random Forest model achieved better performance compared to other algorithms
- Visualizations helped in understanding churn trends and model results
- Early churn prediction enables proactive customer retention strategies
- The proposed system supports data-driven decision making

Pending Taks from Project after Review-1

Sno	Task	Date of Completion
1	Data Preprocessing and Feature Engineering	
2	Model Training and Performance Evaluation	

Thank You

(DEEMED TO BE UNIVERSITY)